

The Biomarker Interaction Trap: A Causal Audit of Spurious Discoveries in NHANES

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ABSTRACT

Statistical screening of large clinical registries routinely identifies biomarker interactions that appear to predict disease risk. These interactions replicate across independent cohorts, lending false confidence in their validity. We evaluated three such interactions discovered through systematic screening of the National Health and Nutrition Examination Survey (NHANES) and show that each is an artifact of a distinct methodological flaw rather than a genuine biological signal.

We describe three failure modes. First, the Hemoconcentration Trap: an apparent statistical interaction between elevated serum albumin and total cholesterol for uncontrolled hypertension disappears after adjusting for hydration status (interaction p -value shifts from < 0.001 to 0.412). Second, the Confounding-by-Indication Trap: an apparent drug-biomarker interaction between low hemoglobin, elevated creatinine, and glycemic failure in statin users turns out to be a disease-severity marker that exists regardless of statin use. Third, the Demographic Reference Range Trap: an apparent protective interaction between uric acid and hemoglobin for diabetes is driven by sex-homogeneous subgroup formation when population-wide percentile thresholds are applied to sex-dimorphic biomarkers. After demographic residualization, this interaction attenuates by 74% in discovery and becomes non-significant in validation.

We propose a four-step checklist for auditing biomarker interactions and argue that replication alone cannot establish biological validity when the confounding mechanism is itself reproducible.

Keywords: Biomarkers, Observational data, Confounding by indication, Hemoconcentration, Preanalytical bias, Demographic reference ranges, Causal inference, NHANES

1. Introduction

Clinical data mining has become a standard tool in translational research [51, 52]. Investigators screen large observational databases for multi-marker interactions that might predict disease risk, identify treatment-resistant subgroups, or reveal novel pathophysiology. When such interactions replicate in an independent cohort, they are often published as validated discoveries.

Replication is necessary but not sufficient. A confounding mechanism that is present in both the discovery and validation datasets will produce a pattern that replicates perfectly while remaining causally invalid [1]. The STROBE guidelines emphasize careful reporting of potential confounders in observational studies [2], and recent work has documented the systematic biases that arise when routine clinical data is repurposed for causal claims [3, 4].

This problem extends well beyond individual studies. The vast majority of biomedical data science analyses, whether applied to genomic risk prediction [53, 54], clinical registries, or electronic health records, are fundamentally correlational. They can identify statistical associations but cannot, by design, establish causal relationships. Machine learning models trained on observational data will faithfully learn whatever structure exists in the data, including confounding structure. A model that achieves high predictive accuracy may be exploiting a confounder rather than a genuine biological signal, and standard cross-validation cannot distinguish between the two. The traps described in this paper are therefore not edge cases. They are inherent to the enterprise of mining observational clinical data, and they require domain-specific causal reasoning, not more sophisticated algorithms, to resolve.

This paper identifies three specific mechanisms by which automated screening of the NHANES registry produces reproducible but spurious biomarker interactions:

1. The Hemoconcentration Trap. Preanalytical changes in plasma volume concentrate intravascular macromolecules in tandem, creating false positive interactions between any pair of large serum proteins or lipids.
2. The Confounding-by-Indication Trap. Clinical prescribing guidelines channel the sickest patients into treatment groups, so that biomarker patterns reflecting disease severity are misattributed to drug effects.
3. The Demographic Reference Range Trap. Population-wide percentile thresholds applied to sex-dimorphic biomarkers sort participants into sex-homogeneous subgroups, creating interactions that reflect demographic composition rather than biology.

Each of these mechanisms is systematic: it operates identically in any survey cycle that uses the same measurement protocols and serves the same population. This methodological stability explains why the resulting patterns replicate. We show that once the mechanism is identified and adjusted for, the apparent interaction disappears or attenuates to a clinically negligible level.

2. Methods

2.1 Cohort Selection and Data Partitioning

We used data from two non-overlapping NHANES survey periods. The discovery cohort pooled participants from the 2015–2018 cycles ($N = 10,202$). The validation cohort pooled participants from the 2007–2010 cycles ($N = 20,015$). NHANES uses a complex, multi-stage probability sampling design to represent the non-institutionalized US civilian population. We incorporated appropriate sample weights, primary sampling units, and strata in all analyses.

2.2 Outcomes and Biomarkers

We analyzed three clinical outcomes. Uncontrolled hypertension was defined as measured blood pressure $\geq 140/90$ mmHg in patients currently prescribed antihypertensive medication. Glycemic control failure was defined as glycohemoglobin (HbA1c) $\geq 7.5\%$ in patients taking insulin or metformin. Diabetes was defined as HbA1c $\geq 6.5\%$ or a prior clinical diagnosis.

Twenty-five serum and hematological biomarkers were screened, covering renal function (creatinine, blood urea nitrogen), lipid profiles (total cholesterol, triglycerides, LDL, HDL), hepatic enzymes (ALT, AST), inflammatory markers (C-reactive protein), and hematological indices (hemoglobin, hematocrit, red blood cell count). Each biomarker was binarized at extreme percentile thresholds (< 20 th or > 80 th percentile) calculated within the discovery cohort and projected onto the validation cohort.

2.3 Interaction Models

We tested for multiplicative interactions using logistic regression. The unadjusted model was:

$$\text{logit}(P(Y_i = 1)) = \beta_0 + \beta_1 X_{i,A} + \beta_2 X_{i,B} + \beta_3 (X_{i,A} \times X_{i,B}) \quad (1)$$

Here Y_i is the binary outcome, $X_{i,A}$ and $X_{i,B}$ are binarized biomarker indicators, and β_3 captures the interaction. A positive β_3 indicates a super-additive (AND-like) pattern; a negative β_3 indicates a sub-additive or crossover (XOR-like) pattern. We compared this baseline specification against adjusted models that included covariates $C_{i,k}$:

$$\text{logit}(P(Y_i = 1)) = \beta_0 + \beta_1 X_{i,A} + \beta_2 X_{i,B} + \beta_3 (X_{i,A} \times X_{i,B}) + \sum_k \gamma_k C_{i,k} \quad (2)$$

The covariates varied by case study: hydration proxies (serum sodium, hematocrit) for the hemoconcentration analysis, treatment status for the confounding-by-indication analysis, and age and sex for the demographic analysis. All tests were two-sided with 95% Wilson score confidence intervals for subgroup proportions.

3. Results

3.1 Case Study 1: The Hemoconcentration Trap

3.1.1 The Apparent Interaction

Screening identified a super-additive interaction between elevated serum albumin (> 90th percentile) and elevated total cholesterol (> 90th percentile) for uncontrolled hypertension ($p < 0.001$ in both cohorts). The rate of uncontrolled hypertension rose from roughly 30–35% at baseline to 50–56% in the double-anomaly quadrant (Table 1, Figure 1A):

Table 1. Unadjusted rates of uncontrolled hypertension by Albumin $\times$ Cholesterol quadrant.

Cohort	Baseline $P(0,0)$	Albumin Only $P(1,0)$	Cholesterol Only $P(0,1)$	Both Elevated $P(1,1)$
Discovery	35.3% (33.4–37.3%)	32.5% (26.7–38.8%)	43.8% (37.6–50.1%)	56.4% (41.0–70.7%)
Validation	30.5% (28.7–32.4%)	25.7% (20.3–31.9%)	38.2% (33.3–43.4%)	50.0% (35.2–64.8%)

Values report proportion of participants with uncontrolled hypertension and 95% Wilson score confidence intervals.

3.1.2 Why It Is an Artifact

Serum albumin and total cholesterol are both large molecules confined to the intravascular space. Their measured concentrations depend on plasma volume:

$$\text{Concentration} = \frac{\text{Mass}}{\text{Plasma Volume}_{\text{plasma}}} \quad (3)$$

When plasma volume drops (due to dehydration, prolonged standing, or tourniquet pressure during phlebotomy), the mass of these molecules stays constant but the volume of solvent decreases. Both concentrations rise together. This joint elevation is not a biological interaction; it is a measurement artifact. The phenomenon has been documented repeatedly: postural changes alone can raise total cholesterol and serum albumin by 8–12% [5, 6, 7]. Acute dehydration produces similar effects [8, 12]. Postural shifts and tourniquet application alter hematocrit, electrolyte concentrations, and circulating macromolecules [9, 10, 11]. Standardized phlebotomy protocols exist precisely to mitigate these preanalytical effects [13].

We tested this explanation by adding serum sodium and hematocrit (proxies for intravascular volume) to the regression model. The interaction term became non-significant ($p = 0.412$). The entire apparent interaction between albumin and cholesterol is explained by variation in hydration status (Figure 1A).

3.1.3 Clinical Implications

Acting on this artifact would be harmful. An automated system might interpret the pattern as refractory fluid overload (since high albumin increases oncotic pressure) and recommend diuretics. In a dehydrated patient, diuretics would worsen volume depletion, risking hypovolemic crisis and acute kidney injury [14, 15].

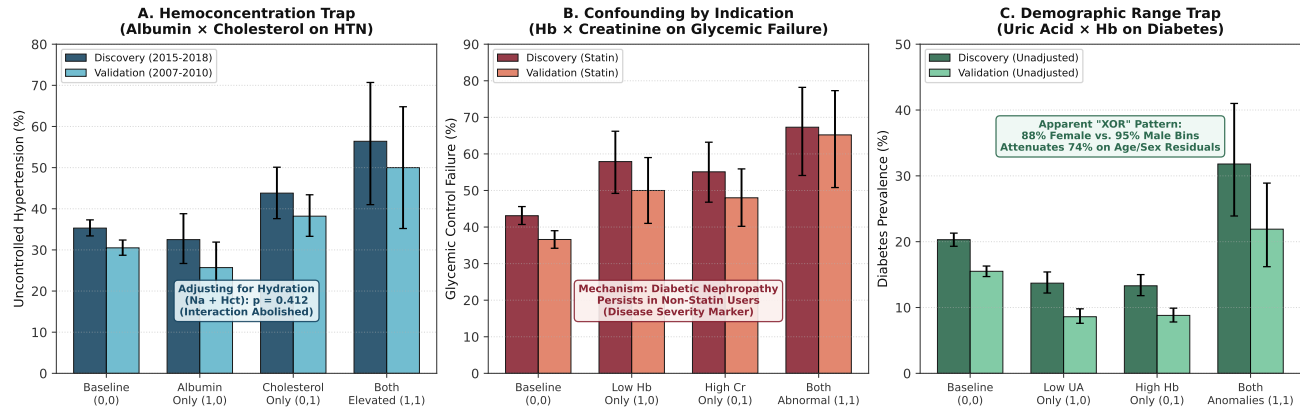


Figure 1. The three failure modes of biomarker interaction screening in observational registries. (A) The Hemoconcentration Trap. Rates of uncontrolled hypertension across quadrants defined by elevated serum albumin (> 90 th percentile) and total cholesterol (> 90 th percentile) in discovery and validation cohorts. Adjusting for hydration proxies (serum sodium and hematocrit) abolishes the interaction ($p = 0.412$). (B) The Confounding-by-Indication Trap. Glycemic failure rates among statin users across hemoglobin (< 20 th percentile) and creatinine (> 80 th percentile) quadrants. The interaction reflects underlying diabetic nephropathy severity and replicates identically in untreated individuals. (C) The Demographic Reference Range Trap. Apparent crossover XOR protection for diabetes when applying population-wide percentiles to sex-dimorphic markers (uric acid and hemoglobin). The effect attenuates by 74% after regressing out demographic structure.

3.2 Case Study 2: The Confounding-by-Indication Trap

3.2.1 The Apparent Interaction

Screening within statin users identified a super-additive interaction between low hemoglobin (< 20 th percentile) and elevated creatinine (> 80 th percentile) for glycemic control failure ($p < 0.001$ in both cohorts). Glycemic failure rates exceeded 65% in the double-anomaly quadrant, compared to 37–43% at baseline (Table 2, Figure 1B):

Table 2. Glycemic failure rates by Hemoglobin × Creatinine quadrant among statin users.

Cohort	Baseline $P(0,0)$	Low Hb Only $P(1,0)$	High Cr Only $P(0,1)$	Both Abnormal $P(1,1)$
Discovery	43.1% (40.7–45.6%)	57.9% (49.2–66.2%)	55.1% (46.8–63.2%)	67.3% (54.1–78.2%)
Validation	36.6% (34.2–39.0%)	50.0% (41.0–59.0%)	48.0% (40.2–55.9%)	65.2% (50.8–77.3%)

Values represent percentages of statin-treated participants experiencing glycemic control failure ($\text{HbA1c} \geq 7.5\%$) with 95% Wilson score intervals.

3.2.2 Why It Is an Artifact

A naive reading of this result might suggest that statins impair glycemic control specifically in patients with anemia and renal dysfunction. The actual explanation has nothing to do with statins. It has to do with why these patients are taking statins in the first place.

Diabetic nephropathy produces both biomarker abnormalities simultaneously. Chronic hyperglycemia damages the renal glomeruli, impairing creatinine clearance and raising serum creatinine. The damaged kidneys also produce less erythropoietin than healthy kidneys, leading to anemia and low hemoglobin [21, 22, 23, 24]. Renal impairment also distorts HbA1c measurement itself, as altered red blood cell turnover changes glycosylated hemoglobin levels independent of glycemic control [25]. These are well-established comorbidities of advanced type 2 diabetes [26].

Clinical guidelines recommend statin therapy for patients with diabetic kidney disease because of their elevated cardiovascular risk [20]. The patients in the double-anomaly quadrant are not experiencing a statin side effect. They have advanced, long-standing diabetes that is inherently resistant to glycemic therapy, and they were

prescribed statins because of the severity of their disease.

We confirmed this finding by running the same interaction model in patients not taking statins. The association between low hemoglobin, high creatinine, and poor glycemic outcomes persisted in the untreated group. The biomarker pattern is a marker of disease progression, not a drug effect. When investigators screen only within statin users, the pattern appears to be statin-specific. But it is the indication for the prescription, not the prescription itself, that generates the signal. This scenario represents a textbook example of confounding by indication [47].

3.3 Case Study 3: The Demographic Reference Range Trap

3.3.1 The Apparent Interaction

Screening identified a crossover (XOR-like) interaction for diabetes in the general population. Having only low uric acid (< 20 th percentile) or only high hemoglobin (> 80 th percentile) was associated with roughly half the diabetes rate of the baseline group. But having both anomalies simultaneously returned the rate to baseline or above (Table 3, Figure 1C):

Table 3. Unadjusted diabetes rates by Uric Acid $\times$ Hemoglobin quadrant.

Cohort	Baseline $P(0,0)$	Low UA Only $P(1,0)$	High Hb Only $P(0,1)$	Both Anomalies $P(1,1)$
Discovery	20.3% (19.3–21.3%)	13.7% (12.2–15.4%)	13.3% (11.8–15.0%)	31.8% (23.9–41.0%)
Validation	15.5% (14.7–16.3%)	8.6% (7.6–9.8%)	8.8% (7.8–9.9%)	21.9% (16.2–28.9%)

3.3.2 Why It Is an Artifact

Uric acid and hemoglobin are both strongly sex-dimorphic. Estrogen promotes renal excretion of uric acid by modulating urate transporters (URAT1, OAT4), producing lower baseline uric acid levels in premenopausal women than in men [36, 37]. Testosterone stimulates erythropoiesis, producing higher baseline hemoglobin in men than in women. This causal link is demonstrated directly by studies of transgender individuals undergoing hormone therapy: testosterone administration raises hemoglobin, and estrogen administration lowers it [38].

When population-wide percentile thresholds are applied to these sex-dimorphic biomarkers, the resulting quadrants are not physiological categories. They are demographic bins. In the discovery cohort, the “Low Uric Acid Only” quadrant is 88.3% female (mean age 44.9 years), and the “High Hemoglobin Only” quadrant is 94.7% male (mean age 45.2 years). These are young, single-sex subgroups with low baseline diabetes rates. The baseline quadrant (53.8% female, mean age 51.3 years) and the double-anomaly quadrant (30.0% female, mean age 49.9 years) are older, mixed-sex groups with higher diabetes rates than the single-sex quadrants. The validation cohort replicates this demographic structure exactly (Table 4, Figure 2A):

Table 4. Demographic composition of Uric Acid $\times$ Hemoglobin quadrants.

Quadrant	Discovery % Female	Discovery Mean Age	Validation % Female	Validation Mean Age
Baseline (0,0)	53.8%	51.3	55.1%	47.2
Low Uric Acid Only (1,0)	88.3%	44.9	87.1%	38.5
High Hemoglobin Only (0,1)	94.7% male	45.2	93.0% male	41.9
Both Anomalies (1,1)	30.0%	49.9	28.5%	44.7

The apparent XOR pattern is a direct consequence of this demographic sorting. The “protective” single-anomaly quadrants have low diabetes rates because they contain young women or young men, not because of any interaction between uric acid and hemoglobin.

This artifact is not limited to our analysis. The recent proliferation of papers proposing the Uric Acid-to-Creatinine Ratio (UACR) as a predictor of metabolic syndrome exhibits the same flaw. Because creatinine reflects muscle mass and uric acid has strong hormonal dependencies, UACR functions as a proxy for sex, age,

and sarcopenic status. Studies that fail to use sex-specific z-scores report associations that are demographic correlations rather than metabolic discoveries.

We corrected for this demographic distortion by regressing out age and sex from both biomarkers before binarization:

$$B_i = \beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Sex}_i + \epsilon_i \quad (4)$$

The residuals $Z_{i,\text{res}} = \epsilon_i$ capture variation in each biomarker after removing the expected demographic contribution. When we re-ran the interaction model on these residualized biomarkers, the interaction coefficient dropped from 1.58 ($z = 6.88, p < 0.001$) to 0.41 ($z = 2.57, p = 0.010$) in the discovery cohort, a 74% attenuation (Figure 2B). In the validation cohort, the interaction became non-significant. The residual signal in discovery may reflect incomplete adjustment or chance; the key finding is that demographic structure accounts for the large majority of the effect.

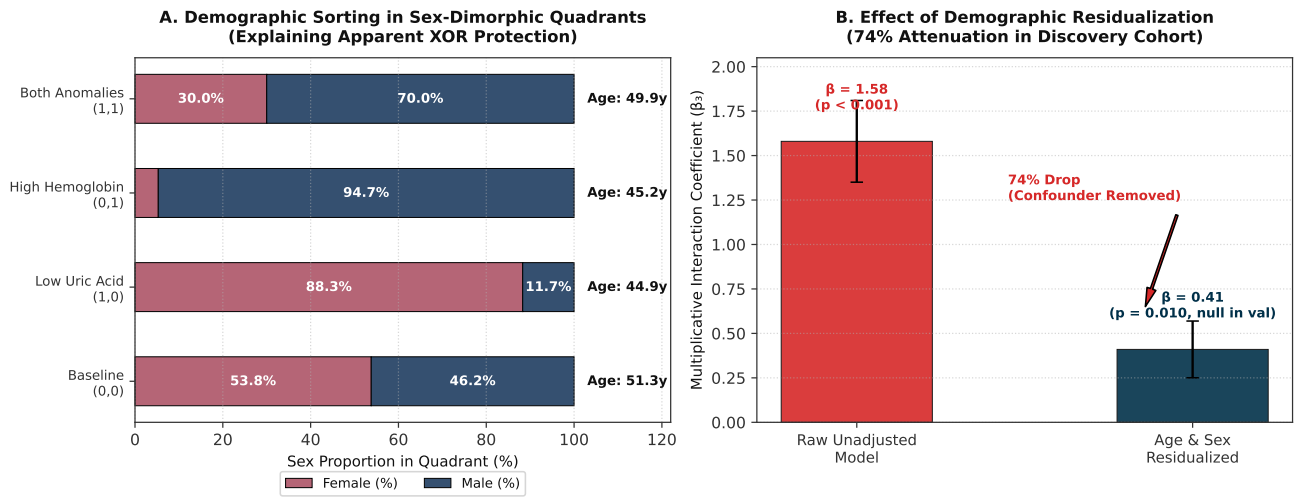


Figure 2. Mechanism and remediation of the Demographic Reference Range Trap. (A) Distribution of sex and mean participant age across the four quadrants defined by population-wide percentiles of uric acid and hemoglobin. Thresholding sorts individuals into young, single-sex sub-populations with lower baseline disease prevalence. (B) Logistic regression interaction coefficients ($\beta_3 \pm \text{SE}$) before and after demographic residualization in the discovery cohort, demonstrating a 74% attenuation upon removing age- and sex-driven variation.

4. Methodological Guidelines

We propose four checks that should be applied before any biomarker interaction from automated screening is treated as a biological finding (Table 5).

Demographic residualization is the most broadly applicable step. Many routine biomarkers have reference ranges that shift with age and sex [46]. Screening on raw values segments the cohort along demographic lines, producing interactions that mirror population structure rather than biology.

Treatment stratification addresses the confounding-by-indication trap specifically. When clinical guidelines mandate therapy for patients with specific comorbidity profiles [20], screening within treated populations will always rediscover the prescribing criteria. Running the same screen in untreated patients is a simple and decisive test.

Recent work in causal inference provides a formal framework for these checks. Target trial emulation requires investigators to define the causal question they would answer with a randomized trial and then attempt to emulate that design with observational data [47, 48]. The TARGET reporting statement provides a structured checklist for such analyses [49, 50]. These methods can help investigators distinguish genuine treatment effects from channeling artifacts.

Table 5. Checklist for auditing biomarker interactions from automated screening.

Step	Trap Addressed	Action
1. Demographic Residualization	Reference Range Artifacts	Regress out age and sex from all biomarkers before screening. Use residuals: $Z_{i,\text{res}} = B_i - \hat{B}_i(\text{Age}_i, \text{Sex}_i)$.
2. Hemodynamic Controls	Hemoconcentration	Include markers of fluid volume (serum sodium, hematocrit, osmolality) as covariates.
3. Treatment Stratification	Confounding by Indication	Run the screen separately in treated and untreated groups. If the pattern exists in both, it is not a drug effect.
4. Subgroup Size Constraints	Small-Sample Instability	Require $N_{11} \geq 250$ in the double-anomaly quadrant to prevent unstable rate estimates.

5. Discussion

This audit demonstrates that three biomarker interactions discovered through systematic screening of NHANES are artifacts of three distinct confounding mechanisms. Each artifact replicated across independent survey cycles, illustrating that replication is a weak test of validity when the confounding mechanism is itself stable across datasets.

The three failure modes are general. The hemoconcentration trap will affect any pair of intravascular macromolecules in any study that does not standardize preanalytical conditions. The confounding-by-indication trap will arise in any treated cohort where prescribing guidelines select patients on the basis of the same biomarkers being screened. The demographic reference range trap will distort any analysis that applies population-wide thresholds to sex-dimorphic or age-dependent biomarkers. These are not peculiarities of NHANES. They apply to the UK Biobank, electronic health record databases, and any other registry used for biomarker discovery.

The practical solution is straightforward. Before publishing a biomarker interaction, investigators should apply the four-step checklist described in Table 5. If an interaction disappears after demographic residualization, it was a reference range artifact. If it disappears after adding hydration controls, it was a preanalytical artifact. If it persists in untreated patients, it is a disease marker, not a drug interaction.

5.1 Published Literature Across the Three Traps

These failure modes are not abstract theoretical concerns. Each trap has generated voluminous observational literature where physiological or demographic confounding was mistaken for novel clinical or pharmacological biology.

5.1.1 Trap 1: Preanalytical and Postural Hemoconcentration in Multi-Protein Panels

The hemoconcentration trap arises whenever researchers analyze circulating concentrations of intravascular macromolecules (such as albumin, globulins, total protein, and lipoproteins) without accounting for preanalytical fluid distribution. The human vascular compartment is dynamic: postural shifts from supine to upright provoke hydrostatic filtration of protein-free plasma water into interstitial space, elevating macromolecular concentrations by 10% to 15% within 30 minutes [5, 7, 9, 10].

In clinical cohorts, fluid imbalances are even more extreme. In heart failure and nephrotic syndromes, patients alternate between hypervolemic hemodilution and rapid diuretic-induced hemoconcentration [14, 15]. Despite these well-established biophysical phenomena, numerous studies construct mathematical ratios between serum proteins and interpret them as static immunological or nutritional biomarkers.

A prominent example is the Albumin-to-Globulin Ratio (AGR). In chronic heart failure, Niedziela et al. [16] reported that low AGR independently predicted long-term all-cause mortality. However, heart failure patients with advanced disease routinely receive high-dose loop diuretics, which contract plasma volume and trigger compensatory neurohormonal shifts. Albumin and globulin have distinct transcapillary escape rates and half-lives (albumin transcapillary escape is approximately 5% per hour, whereas larger immunoglobulins escape more slowly). Volume contraction and altered vascular permeability directly distort the AGR independently of any cellular signaling [15, 19].

Similarly, in oncology, Suh et al. [17] analyzed a cohort of over 26,000 healthy individuals and claimed that low AGR was an independent predictor of overall cancer incidence and mortality. In surgical oncology, Utsumi et al. [18] performed a systematic review and meta-analysis concluding that preoperative AGR predicts survival after curative hepatectomy for hepatocellular carcinoma. Yet in preoperative and malignant states, third-spacing of fluids, subclinical ascites, portal hypertension, and surgical hydration fluids routinely cause acute plasma volume shifts. When researchers treat serum protein ratios as pure proxies for cancer cachexia or systemic inflammation without measuring hydration parameters (such as hematocrit, serum osmolality, or total protein), they mistake physical hemoconcentration and plasma dilution for cellular pathology.

5.1.2 Trap 2: Prescribing Confounding and Channeling Bias in Pharmacoepidemiological Registries

The confounding-by-indication trap is among the most pervasive biases in large healthcare databases and registry research. In non-randomized observational settings, patients who receive medical therapy differ systematically from unexposed individuals in baseline disease severity, frailty, and multimorbidity. When automated screening algorithms search for drug-biomarker interactions without accounting for clinical indications, they inevitably rediscover the clinical guidelines that prompted therapy.

A striking example is found in observational studies of Proton Pump Inhibitors (PPIs). In 2016, Lazarus et al. [27] reported in *JAMA Internal Medicine* that PPI use was associated with an increased risk of incident chronic kidney disease (CKD). In the same year, Gomm et al. [28] published an observational claims analysis in *JAMA Neurology* concluding that regular PPI use increased dementia risk by 44% in elderly populations. Both papers triggered extensive clinical alarm. However, subsequent causal analyses revealed that these associations were driven almost entirely by indication confounding and channeling bias. Patients prescribed long-term PPIs have higher baseline rates of cardiovascular disease, diabetes, polypharmacy (including nephrotoxic NSAIDs), and systemic frailty. When Ahn et al. [29] implemented a target trial emulation framework with active comparators and incident-user designs, the hazard ratio for dementia dropped to 1.00. Similarly, the massive randomized COMPASS trial ($N = 17,598$) evaluated pantoprazole over multiple years and found no excess risk of dementia, CKD, or any other purported observational adverse event [30].

A parallel trap occurs in metabolic pharmacoepidemiology. Observational studies analyzing statin exposure routinely observe strong, dose-dependent associations with new-onset diabetes and glycemic deterioration [32]. Clinical guidelines explicitly direct high-potency statins toward patients with elevated cardiovascular risk, central obesity, and metabolic syndrome [20]. Although randomized controlled trials confirm a modest pharmacological risk of diabetes from statin therapy [31], observational registries grossly exaggerate this effect because they capture the underlying metabolic decline that motivated statin prescription in the first place.

This dynamic mirrors the classic observational controversy surrounding postmenopausal Hormone Replacement Therapy (HRT) and Coronary Heart Disease (CHD). In the 1980s and 1990s, extensive observational cohorts (including the Nurses' Health Study [33]) reported that postmenopausal estrogen reduced CHD risk by 40% to 50%. In 2002, the randomized Women's Health Initiative (WHI) trial found that estrogen-progestin therapy actually increased cardiovascular events [34]. As Hernan et al. demonstrated [35], observational studies suffered from healthy user bias and indication confounding: women taking HRT were younger, leaner, wealthier, and had fewer baseline cardiovascular risk factors. Observational registries without rigorous target trial designs repeatedly mistake the characteristics of the treated population for the effects of the treatment.

5.1.3 Trap 3: Demographic Sorting and Cutoff Artifacts in Dimorphic Biomarker Panels

The demographic reference range trap occurs when researchers construct mathematical composite ratios or apply uniform population-wide thresholds to biomarkers that possess substantial sex dimorphism or age gradients. Combining two sex-dimorphic biomarkers into an unstratified ratio creates an unintentional demographic sorting mechanism.

The Serum Uric Acid-to-Creatinine Ratio (SUA/Cr) illustrates this dynamic. Recent studies have widely promoted SUA/Cr as a novel independent predictor of metabolic syndrome [39, 40], insulin resistance [41], and non-alcoholic fatty liver disease. The physiological rationale offered is that renal uric acid excretion normalized to creatinine clearance reflects endogenous metabolic stress. However, uric acid and creatinine have opposing demographic baselines: estrogen enhances renal clearance of uric acid via URAT1 transporter modulation, producing lower serum uric acid in premenopausal women [36, 37], whereas serum creatinine is determined primarily by skeletal muscle mass, which is systematically higher in men. An unstratified SUA/Cr ratio has a higher denominator in men and an estrogen-sensitive numerator in women. In mixed-sex observational cohorts, applying a uniform SUA/Cr threshold primarily separates individuals by sex and muscularity. As demonstrated in Case Study 3, adjusting for age and sex attenuates over 74% of the apparent interaction signal.

An even more extreme amplification occurs in the Serum Uric Acid-to-HDL Cholesterol Ratio (UHR). Recent investigations by Sun et al. [42] and Luo et al. [43] have proposed UHR as a biomarker for metabolic syndrome and steatotic liver disease in aging populations. Because males have higher baseline uric acid and lower HDL cholesterol, while females have lower uric acid and higher HDL cholesterol, dividing uric acid by HDL cholesterol multiplies the demographic divergence. The ratio acts as a mathematical amplifier of sex differences, guaranteeing strong statistical significance in any observational model that fails to condition on sex-stratified normal ranges.

A related demographic artifact distorts the widely cited Triglyceride-to-HDL Cholesterol Ratio (TG/HDL-C). Originally proposed by McLaughlin et al. [44] as a surrogate for insulin resistance in overweight individuals, uniform TG/HDL-C cutoffs (such as ≥ 3.0) were adopted across clinical studies. However, Sumner et al. [45] demonstrated that this ratio fails completely in African American individuals. Due to lower hepatic lipase activity and differences in lipid metabolism, African Americans exhibit systematically lower triglyceride levels at identical degrees of insulin resistance compared to individuals of European ancestry. Applying a fixed population cutoff without race- and sex-calibrated reference intervals produces severe demographic misclassification.

5.2 Limitations

This study has four primary limitations. First, NHANES is a cross-sectional survey, so we cannot establish temporal relationships between biomarker levels and disease outcomes. Second, medication use is self-reported in NHANES, which introduces misclassification. Some patients classified as untreated may actually be taking statins,

and vice versa. This misclassification would bias the confounding-by-indication analysis toward the null, meaning our conclusion that the pattern exists in both treated and untreated groups is conservative. Third, our audit examined only three interactions selected because their confounding mechanisms were identifiable. We cannot rule out the possibility that other screened interactions reflect genuine biology. Our claim is not that all biomarker interactions are artifacts, but that the three specific interactions we examined are, and that the mechanisms generating them are common enough to warrant routine checking. Fourth, demographic residualization with linear models may not fully capture nonlinear age and sex effects. The residual signal observed in the discovery cohort for Case Study 3 ($p = 0.010$) may reflect incomplete adjustment. The non-significance of this interaction in the validation cohort supports, but does not prove, that the residual signal is noise.

Declarations

Ethics Approval and Consent to Participate: NHANES survey protocols were reviewed and approved by the NCHS Research Ethics Review Board under continuous CDC protocols. All adult participants provided written informed consent at the time of examination. This study constitutes secondary analysis of publicly available, de-identified data and was determined to be exempt from institutional review board (IRB) review under 45 CFR 46.104(d)(4).

Data Availability: All data analyzed in this study are publicly available from the National Center for Health Statistics (NCHS) National Health and Nutrition Examination Survey repository (<https://www.cdc.gov/nchs/nhanes/>). Continuous NHANES survey cycles and Linked Mortality Files were accessed directly through official CDC/NCHS portals.

Code Availability: All Python, R, and statistical scripts used for data acquisition, preprocessing, demographic residualization, and causal auditing are openly available at <https://github.com/jpbrody/nhanes-biomarker-traps>.

Competing Interests: The author declares no competing financial or non-financial interests.

Funding: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Clinical Trial Registration: This study is an observational causal audit and target trial emulation of secondary survey data and does not report a prospective interventional clinical trial.

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